
Cloud Operations

Module 1: Introduction to Cloud Operations and AWS

Fundamentals

July 17, 2025

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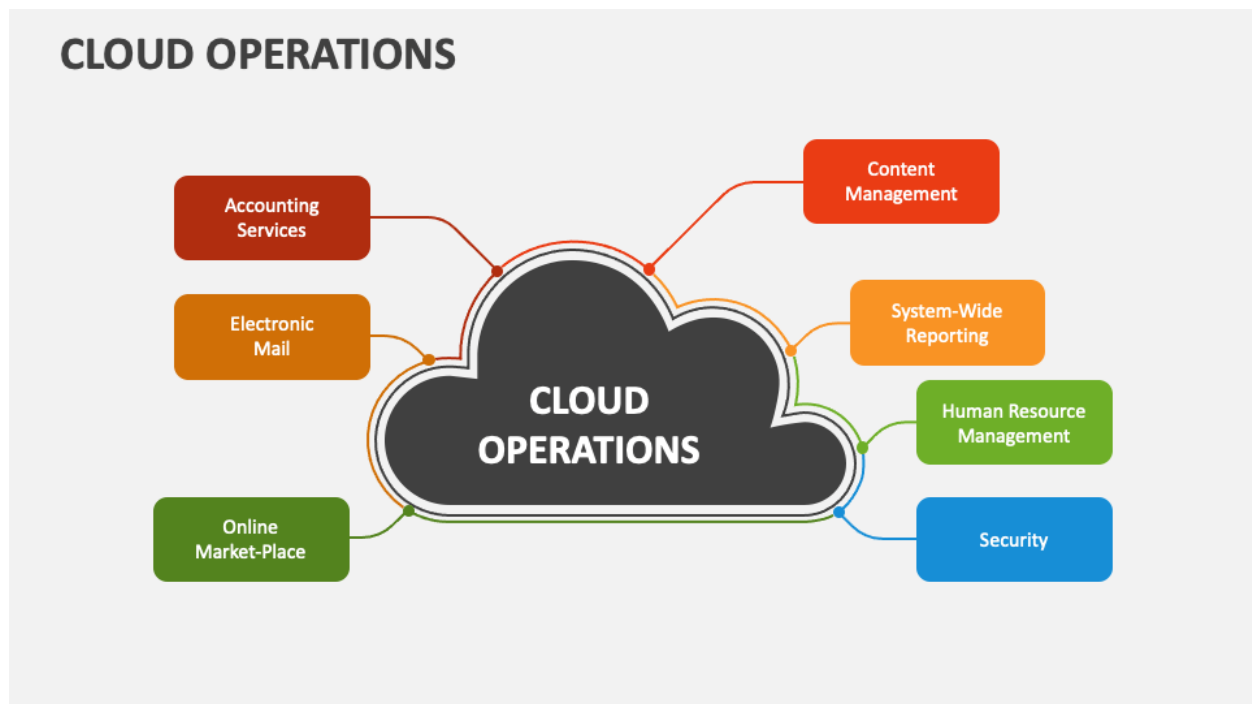
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1. Cloud Operations (CO)

1.1 Introduction

It is defined as the environment of cloud operations in which you will see how you can identify and defining appropriate or standards for IT services. It is very helpful to build the pipeline for IT operations and optimizing cloud operations. It includes network combinations, maintaining security, providing performance, proper device management, help desk, and some other tasks that are used by cloud applications and various cloud operations.



1.2 Features

- CloudOps acts as a key partner collaborator in defining and accelerate the cloud-native journey.
- It helps to achieve the targeted goals in business while deploying, building, and delivering cloud services.
- Moreover, it helps hundreds of enterprises and web-based companies to build their business in the cloud without scarifying any performance, compliance, cost, and privacy.
- It is specific management, delivery, and consumption of software in a computing environment where there is limited visibility into apps infrastructures.

- In various enterprises, this CloudOps uses DevOps principles of Continuous Integration (CI) and Continuous Deployment (CD) to create some best practices for achieving better business processes.

1.3 Responsibilities

1. It is responsible for Configuration management.
2. Resources allocation is one of the keys responsibilities where resources will be allocated on the basis of cloud operations.
3. Using CloudOps is one of the main objectives to Optimizes performance capacity.
4. To Manages compliance burden on the basis of different cloud-based operations.
5. It Ensures service level agreements.

1.4 Foundation

The keys concepts of the CloudOps foundation are abstraction, automated, provisioning, policy, and process automation. Whereas the abstraction decouples management from underlying infrastructure in instances, security, network, and governance can be managed through a single pane.

1.5 Functions

The main function of CloudOps is to organize teams within the required skills sets which also includes IT operations and their teams, DevOps teams, business application operations, and cloud services.

1. Keeps the cloud services and infrastructure up and ongoing.
2. Optimizes performance and capacity in a cost-effective way.
3. Fulfill all the service level agreements
4. Self-service cloud access
5. Works on both compliance and configuration loops.
6. Does proper recovery on time

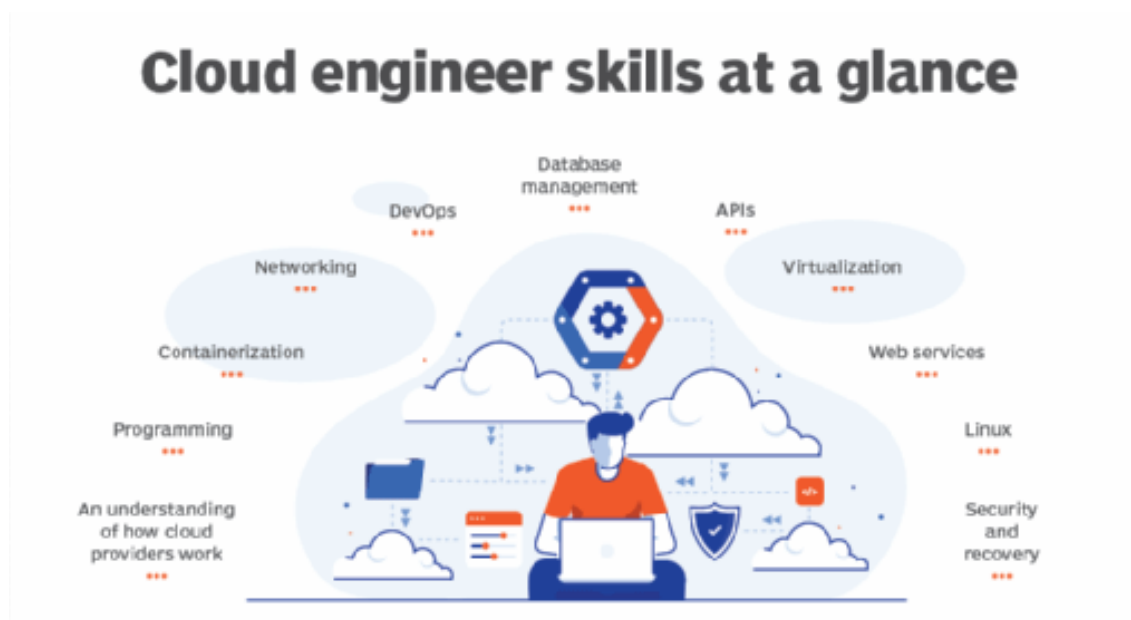
1.6 Management

- Cloud operation is easy but only when the management is done in a superior way. As observed the organized cloud resources absolutely revolutionize the operation.

- Simply CloudOps can't be organized by introducing or merging automatically it with the pre-existing DevOps system. The process compilation and enforcement in between two are needed to be balanced by their respective teams.
- Some basic strategies to follow up are – preparation, creating a cloud checklist, finding the best organization, visualizing network and infrastructure, investing in tools, keep up with maintenance requirements.

2.Roles and Responsibilities of CO Engineer

A cloud operations engineer is a professional responsible for managing, monitoring and optimizing cloud-based infrastructure, applications and services within an organization. They play a crucial role in ensuring the smooth functioning of cloud environments by handling tasks such as resource provisioning, configuration management, deployment, automation and incident response.



Additionally, they collaborate with development and IT teams to design and implement scalable, secure and cost-effective cloud solutions that align with business objectives and end-user requirements. Cloud operations engineers must possess a deep understanding of cloud computing technologies, platforms and best practices, as well as strong problem-solving, communication and collaboration skills.

A CloudOps engineer plays a vital role in managing and maintaining cloud infrastructure and services. Their responsibilities and roles might include:

- **Cloud infrastructure management:** Design, deploy and maintain cloud infrastructure, ensuring optimal performance, security and scalability.
- **Cloud migration:** Responsible for planning, executing and managing cloud migration of applications, workloads and infrastructure from on-premises or other cloud environments to a target cloud platform.
- **Resource provisioning:** Allocate and manage cloud resources such as compute, storage and networking to meet the needs of applications and services.
- **Configuration management:** Define and implement infrastructure configurations and manage changes to maintain consistency across environments.
- **Automation:** Develop and implement automation scripts and tools to streamline and simplify repetitive tasks, improving efficiency and reducing human error.
- **Monitoring:** Set up and maintain monitoring tools to track the performance, availability and security of cloud services, proactively identifying and resolving issues.
- **Incident response:** Troubleshoot and resolve incidents, collaborating with development and IT teams to minimize downtime and maintain service quality.
- **Security and compliance:** Implement and enforce security policies, best practices and compliance requirements to protect sensitive data and maintain cloud infrastructure integrity.
- **Cost optimization:** Monitor and analyze cloud resource usage to identify opportunities for cost savings, recommending and implementing cost optimizations.
- **Backup and disaster recovery:** Develop and maintain backup and disaster recovery plans to ensure data and application availability during unforeseen events.
- **Collaboration and communication:** Work closely with development, IT and business teams to ensure alignment with project goals and provide guidance on cloud best practices.
- **Continuous improvement:** Stay up to date with emerging cloud technologies, platforms and trends, continuously improving cloud operations and adapting to changing requirements.

- **Documentation:** Create and maintain comprehensive documentation of cloud infrastructure, configurations, processes and procedures to ensure knowledge sharing and team collaboration.

3. Overview of Cloud Operating Environments

Cloud operating environments provide the infrastructure and platform for delivering computing services over the internet, allowing users to access resources on demand without managing physical hardware. These environments are characterized by on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service. They encompass various models like public, private, and hybrid clouds, with services like IaaS, PaaS, and SaaS.

3.1 Key Aspects

- **On-demand self-service:** Users can provision resources and services as needed, without requiring human intervention.
- **Broad network access:** Services are accessible over standard network protocols from various devices.
- **Resource pooling:** Multiple users share underlying resources, allowing for efficient utilization and cost optimization.
- **Rapid elasticity and scalability:** Resources can be scaled up or down quickly to meet fluctuating demands.
- **Measured service:** Usage of resources is tracked and metered, allowing for pay-as-you-go pricing.
- **Virtualization:** Cloud environments rely heavily on virtualization to abstract hardware resources and enable efficient resource allocation.
- **Automation:** Cloud operations leverage automation tools for tasks like infrastructure provisioning and configuration management.
- **Multi-tenancy:** Cloud resources can be shared by multiple users or organizations simultaneously, enhancing efficiency and cost-effectiveness.

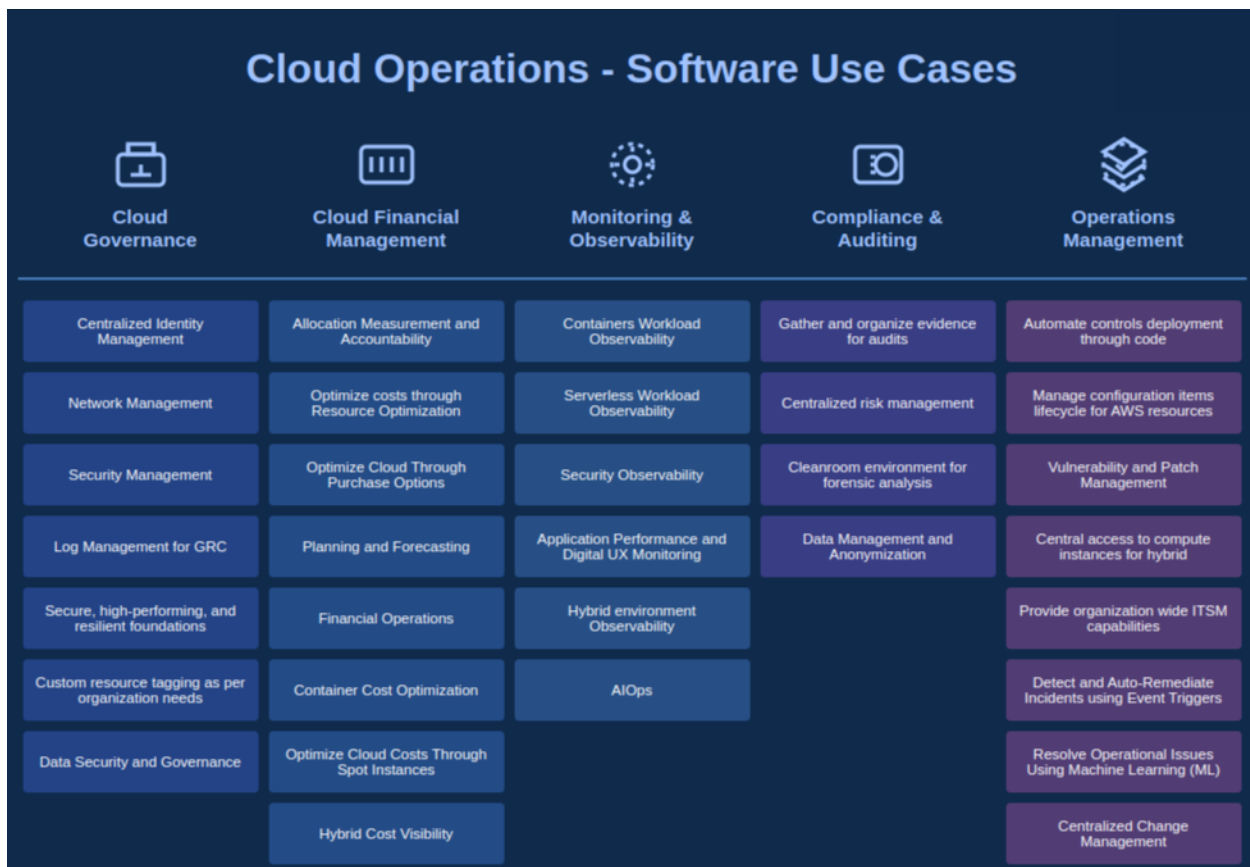
3.2 Types

- **Public Cloud:** Services are provided over the public internet by a third-party provider and are accessible to anyone.

- **Private Cloud:** A cloud environment dedicated to a single organization, offering enhanced control and security.
- **Hybrid Cloud:** A combination of public and private clouds, allowing for flexible resource allocation and workload management.
- **Multi-cloud:** Utilizing multiple public or private cloud services, often to leverage different strengths or optimize costs.

3.3 Cloud Services Models

- **Infrastructure as a Service (IaaS):** Provides virtualized computing resources like servers, storage, and networking.
- **Platform as a Service (PaaS):** Offers a platform for developing, running, and managing applications without managing the underlying infrastructure.
- **Software as a Service (SaaS):** Delivers software applications over the internet, with the provider managing the infrastructure and maintenance.
- **Serverless Computing:** Allows developers to run code without managing servers, abstracting away infrastructure management.



4. Importance of CloudOps in Modern IT Infrastructure

Cloud operations, or CloudOps, are crucial in modern IT infrastructure for enhancing agility, performance, security, and cost efficiency. CloudOps teams manage and optimize cloud-based workloads, ensuring smooth operations, security, and effective resource utilization.

Here's a breakdown of CloudOps' importance:

- **Enhanced Agility and Responsiveness:** CloudOps enables rapid deployment of applications and services, allowing businesses to quickly adapt to market changes and customer demands.
- **Improved Performance:** By monitoring real-time performance and automating tasks, CloudOps ensures optimal performance and user experience.
- **Scalability:** CloudOps facilitates dynamic scaling of resources based on workload demands, ensuring efficient resource utilization and avoiding over-provisioning.
- **Cost Efficiency:** By optimizing resource usage and leveraging pay-as-you-go models, CloudOps helps reduce operational costs and eliminate unnecessary expenses.
- **Strengthened Security:** CloudOps implements security measures like malware protection, data encryption, and incident response plans to protect against threats and ensure data security.
- **Improved Resilience and Disaster Recovery:** CloudOps utilizes automated backups and recovery strategies to ensure business continuity and minimize downtime in case of disruptions.
- **Automation:** CloudOps leverages automation tools for various tasks like provisioning, configuration management, and incident response, improving efficiency and reducing manual errors.
- **Data Mobility and Accessibility:** Cloud operations enable seamless data access from anywhere with an internet connection, supporting remote work and collaboration.
- **Innovation and Faster Time to Market:** CloudOps supports agile development practices, enabling faster development cycles and quicker product releases.

- **Focus on Core Business Activities:** By handling the complexities of cloud infrastructure management, CloudOps allows IT teams to focus on strategic initiatives and innovation.
- **Cloud-Native Security:** CloudOps integrates security best practices into every stage of cloud operations, ensuring a robust security posture.
- **Compliance:** CloudOps helps organizations meet regulatory requirements and industry standards related to data security and privacy.

5. Understanding AWS Services

5.1 Introduction

Amazon Web Services (AWS) is a cloud platform offered by Amazon that lets people and companies use IT services like storage, servers, and databases through the internet. Instead of buying and maintaining physical computers, you can "rent" these services online and only pay for what you use.

Whether you want to host a website, store files, run applications, or analyze large data, AWS provides tools to do it easily and quickly. It offers over 200 services such as EC2 for virtual servers, S3 for storing files, RDS for managing databases, and Lambda to run code without servers. Big companies like Netflix, Airbnb, and NASA use AWS, but it is also perfect for small startups and individuals. It helps you start small, grow as needed, and stay secure with reliable global support.

5.2 Features

1. On-Demand Resources

With AWS, you don't need to set up your computers or servers. You can quickly use services like storage, databases, and virtual machines over the internet. It is like turning on a switch, use what you need, when you need it.

2. Pay-as-You-Go Pricing

AWS doesn't ask you to buy anything upfront. You are charged only for the time and the amount of services you use. If you use a server for 2 hours, you only pay for 2 hours. This helps save money and avoid waste.

3. Global Infrastructure

AWS has data centers across the globe. So, whether your users are in India, the US, or Europe your app can reach them quickly. Also, if your business grows and you need more power or storage, AWS can easily scale up without downtime.

4. Reliability and Performance

AWS is built to be strong and dependable. It uses multiple backup systems, so even if one fails, others keep your data and apps running. You get fast performance and high availability, so your services rarely go down.

5. Everything from one Platform

In AWS, everything you need is available on one platform. You can launch virtual servers using Amazon EC2 to run your applications, store files and media using Amazon S3, and manage databases easily with Amazon RDS. In addition to this, AWS allows you to analyze large amounts of data, build and run artificial intelligence models, and even host complete websites. All these services are accessible from a single dashboard, making it simple to build, deploy, and manage projects without setting up any physical hardware.

5.3 Working

Amazon Web Services (AWS) delivers cloud computing through a network of globally distributed data centers connected by high-speed fiber. These centers host a wide range of services from storage and compute power to databases and machine learning that users can access on demand via the internet.

Whether you are a startup or an enterprise, AWS allows you to build and run applications without managing physical servers. It handles core responsibilities like server maintenance, scaling, patching, and security, so you can focus purely on development and innovation.

For example, if you are running an eCommerce site, instead of manually handling database setup, backups, and failover, you can use **Amazon RDS**. It automates all these backend tasks, enabling you to launch a database in minutes with just a few clicks.

5.4 Fundamentals

In the Journey of AWS, understanding the key concepts such as **Regions, Availability Zones, Global Network Infrastructure, etc** is crucial. The fundamentals of AWS keep on maintaining the applications reliable and scalable with services globally by enabling

strategic deployment of resources for optimal performance and resilience. The following are the some of the main fundamentals of AWS:

- **Regions:** AWS provide the services with respective division of regions. Regions are based on geographical locations, each with one or more data centers. Based on need and traffic of users, the scale of data centers is depended to facilitate users with low-latencies of services.
- **Availability Zones (AZ):** To protect data centers from natural calamities or any other disasters. Data centers are divided into isolated locations called Availability Zones to enhance fault tolerance and disaster recovery management.
- **Global Network Infrastructure:** AWS ensures the reliability and scalability of services through setting up its own AWS Network Infrastructure globally. It helps in better management of data transmissions for optimized performance and security reliance.

5.5 Applications

AWS services are used by both startups and large enterprises based on their specific needs. Startups use AWS to overcome hardware infrastructure costs and deploy applications efficiently. Whereas large scale companies are using AWS cloud services for the management of their Infrastructure to completely focus on the development of products widely.

Here are some real-world use cases of AWS services:

- **Netflix:** The large streaming giant using AWS for the storage and scaling of the applications for ensuring seamless content delivery with low latency without interruptions to millions of users globally.
- **Airbnb:** By utilizing AWS, Airbnb manages the various workloads and provides scalable and reliable infrastructure for its virtual marketplace and lodging offerings.
- **NASA's Jet Propulsion Laboratory:** It takes the help of AWS services to handle and analyze large-scale volumes of data related to vital scientific research missions and space exploration.
- **Capital One:** A financial Company that is utilizing AWS for its security and compliance while delivering innovative banking services to its customers.

5.6 Advantages

Understanding the benefits of AWS helps you see that it is a reliable platform that can support your business needs easily and effectively:

- AWS allows you to easily scale your resources up or down as your needs change, helping you to save money and ensure that your application always has the resources it needs.
- AWS provides a highly reliable and secure infrastructure, with multiple data centers and a commitment to 99.99% availability for many of its services.
- AWS offers a wide range of services and tools that can be easily combined to build and deploy a variety of applications, making it highly flexible.
- AWS offers a pay-as-you-go pricing model, allowing you to only pay for the resources you actually use and avoid upfront costs and long-term commitments.

5.7 Disadvantages

Like any technology, AWS also has its downsides. No system is completely perfect, and AWS comes with a few challenges of its own. Let's take a look at some of them:

- AWS can be complex, with a wide range of services and features that may be difficult to understand and use, especially for new users.
- AWS can be expensive, especially if you have a high-traffic application or need to run multiple services. Additionally, the cost of services can increase over time, so you need to regularly monitor your spending.
- While AWS provides many security features and tools, securing your resources on AWS can still be challenging, and you may need to implement additional security measures to meet your specific requirements.
- AWS manages many aspects of the infrastructure, which can limit your control over certain parts of your application and environment.



6. Overview of AWS Cloud Computing Platform

AWS (Amazon Web Services) is a comprehensive cloud computing platform providing a vast array of on-demand IT resources like compute power, storage, and databases, accessible over the internet. It enables users to build and run applications and services without the need to manage physical infrastructure, offering scalability, flexibility, and cost-effectiveness. AWS is used by a wide range of organizations, from startups to large enterprises and government agencies.

6.1 Key aspects

- **On-demand IT resources:** AWS offers a wide range of services, including compute (virtual machines, serverless functions), storage (object storage, block storage, file storage), databases (relational, NoSQL), networking, analytics, machine learning, and more.
- **Pay-as-you-go pricing:** Users only pay for the resources they consume, making it a cost-effective solution for various needs.
- **Global infrastructure:** AWS has a vast global network of regions and availability zones, ensuring high availability, scalability, and low latency for applications.
- **Security and compliance:** AWS provides robust security features, compliance certifications, and tools to protect data and applications.
- **Flexibility and scalability:** AWS allows users to easily scale resources up or down based on demand, adapting to changing business needs.
- **Innovation and development:** AWS provides tools and services for building and deploying applications, including support for emerging technologies like AI/ML and IoT.
- **Wide adoption:** AWS is used by millions of customers worldwide, including major companies like Netflix, Airbnb, and NASA.

6.2 Popular Services

- **Amazon EC2 (Elastic Compute Cloud):** Provides virtual servers for running applications.
- **Amazon S3 (Simple Storage Service):** Offers object storage for storing and retrieving any amount of data.
- **Amazon RDS (Relational Database Service):** Manages relational databases.

- **Amazon Lambda:** Allows running code without provisioning or managing servers.
- **Amazon VPC (Virtual Private Cloud):** Enables users to create isolated virtual networks in the AWS cloud.
- **Amazon DynamoDB:** A NoSQL database service.
- **Amazon CloudFront:** A content delivery network (CDN) service.
- **Amazon API Gateway:** Allows creation and management of APIs.
- **AWS IAM (Identity and Access Management):** Provides access control and management for AWS resources.

7. Key AWS Services and Their Use Cases

AWS (Amazon Web Services) offers a broad range of services designed to meet the needs of businesses across different sectors. Here, we highlight most important and widely used AWS services.

7.1 Compute Services

1. Amazon EC2 (Elastic Cloud Compute)

- Amazon EC2 is the fastest cloud computing service provided by AWS. It offers virtual, secure, reliable, and resizable servers for any workload. This makes it simple for developers to access resources they need and run large-scale applications.
- EC2 offers the latest processors, networking, and storage options. Developers can quickly increase or decrease server capacity based on business needs.
- It has over 500 types of servers and you can also choose the latest processor, operating system, storage, and networking to help you choose according to the needs of the business. Also, with Amazon EC2, you only have to pay for what you use, and also as per the time period, scale with amazon EC2 auto-scaling has optimal storage and can optimize CPU configurations.

2. Amazon Lambda

- Amazon Lambda is a serverless service from AWS that runs your code automatically when needed. You don't have to manage any servers or clusters. You just upload your code, and Lambda takes care of everything else.
- It's cost-effective because you only pay when your code runs. Lambda can automatically scale and handle many requests at the same time, even thousands

per second. It also makes sure your application is always available. That's why it's one of the best AWS services for developers.

3. Amazon Elastic Beanstalk

- Amazon Elastic Beanstalk is a service from AWS that helps you easily deploy and manage web applications built with Java, PHP, Python, Docker, and more. You just upload your code, and the deployment part is handled by Elastic Beanstalk (from capacity provisioning, load balancing, and auto-scaling to application health monitoring).
- It handles most of the heavy work, but you can still control AWS resources if needed. You only pay for what you use, which helps keep costs low. That's why it's a great choice for developers.

7.2 Database Services

4. Amazon RDS (Relational Database Services)

- Amazon RDS is a managed database service from AWS that supports databases like PostgreSQL, MariaDB, MySQL. With RDS, you can easily create, run, and scale databases in the cloud without managing hardware or installing software.
- It handles tasks like setting up the database, managing servers, updating software, and creating backups. RDS also offers high performance, strong security, and high availability. You can choose from over 15 database engines based on your needs. It's a secure, reliable, and cost-effective service for managing databases.

5. Dynamo DB

- DynamoDB is a serverless NoSQL database from AWS that stores data as key-value pairs and documents. It is made to handle very large, high-speed applications. DynamoDB can process up to 10 trillion requests per day and handle over 20 million requests every second.
- It comes with built-in security, automatic backups, and can copy data across multiple regions for better availability. It's fully managed, meaning AWS takes care of everything for you. DynamoDB also includes tools to analyze data, track traffic, and get useful insights. Features like in-memory caching, easy data import/export, and automatic scaling make it fast, reliable, and easy to use.

7.3 Storage Services

6. Amazon S3 (Simple Storage Service)

- **Amazon S3** is a popular AWS service that makes it easy to store and access data from anywhere at any time. It is a type of object storage that offers high performance, strong security, and can grow as your data grows.
- Data in S3 is stored in different “storage classes,” so you can manage costs without extra investment. It’s a great option for large businesses that need to handle huge amounts of data for different purposes.
- S3 also includes strong access controls, tools to copy data for safety, protection against accidental deletion, and version control to keep track of changes.

7. Amazon EBS (Elastic Block Store)

- **Amazon EBS** is a storage service made for Amazon EC2. It helps safely store and manage data for all kinds of workloads, whether they are small or large.
- You can use EBS for different types of databases, business apps, and big data tools like Hadoop and Spark. It offers five types of storage volumes, so you can pick the one that fits your needs and budget.
- EBS can automatically create and manage backups using its lifecycle management feature. It also supports high-performance applications like Microsoft and SAP and allows you to easily scale as your workload grows.

8. Amazon EFS (Elastic File System)

Amazon EFS (Elastic File System) is a simple and serverless system where you can create and configure file systems without provisioning, deploying, patching, and maintaining. It is a scalable NFS file system made for use in AWS cloud services and on-premises resources. Also, it has no minimum fee or setup charge. You pay for the storage you use such as -

- for provisioned throughput
- automatically expand and shrink as per the addition and removal of files
- read and write access to data stored in Infrequent Access storage classes

It is a scalable service where you can scale up to petabytes without thinking about the performance of the application.

7.4 Networking & Content Delivery

9. Amazon VPC (Virtual Private Cloud)

- **Amazon VPC** is a service that lets you create your own private area inside the AWS cloud. In this private space, you can set up and manage AWS resources safely and on a large scale.
- With VPC, you control how your network works, where resources are placed, how they connect, and how secure they are. You can set rules for incoming and outgoing traffic to improve security.
- VPC can also find unusual activity, fix network issues, prevent data leaks, and handle setup problems. You get full control over the environment, like choosing IP addresses, creating subnets, and managing routing.

10. Amazon CloudFront

- Amazon CloudFront is a service that helps deliver content (like websites, videos, or apps) quickly to people all over the world. It works fast, safely, and with very little delay.
- It uses smart routing and a global network of data centers (called edge locations) to send content from the nearest location to the user. This makes content load faster.
- CloudFront also keeps your data secure with traffic encryption and access controls. It works well with other AWS services like Amazon S3, EC2, and Lambda to handle custom code.
- When used with S3 or EC2, there are no extra charges for data transfer between these services.

11. Elastic Load Balancer (ELB)

- Elastic Load Balancer (ELB) is an AWS service that helps distribute incoming network traffic across multiple servers. This ensures that no single server is overwhelmed with too much traffic and that users get fast and reliable responses.
- ELB helps with high availability, fault tolerance, and the efficient management of traffic in a scalable way. It integrates well with services like EC2, ECS, and Lambda to manage large traffic loads and maintain a seamless user experience.
- ELB fits into cloud environments that require automatic scaling, distributed traffic, and fault tolerance to keep services always available.

7.5 Security & Identity Services

12. Amazon IAM (Identity and Access Management)

- **Amazon IAM** helps you control who can access and use AWS resources safely. It lets you decide who can sign in and what they are allowed to do.
- With IAM, you can create rules based on things like job role or department, so each person only gets access to what they need. You can allow or block access for each user as required.
- IAM acts like a central manager that controls all permissions across AWS, making sure only the right people can use specific tools and resources.

7.6 Monitoring & Analytics

13. Amazon Cloudwatch

- Amazon CloudWatch helps you watch over your AWS environment. It can spot unusual changes, send alerts, fix problems, and even take automatic actions when needed.
- You can monitor your whole system, including applications and resources, by using logs, alarms, and event data. This helps you focus on building your application while CloudWatch handles the monitoring.
- It's a great tool for developers, DevOps engineers, site reliability engineers, and IT managers. CloudWatch checks how well your apps are running, helps improve performance, and makes sure resources are used efficiently.

7.7 Machine Learning & AI

14. Amazon SageMaker

- **Amazon SageMaker** is a machine learning service from AWS. It helps data scientists, business analysts, and developers easily build, train, and deploy machine learning models.
- SageMaker makes data analysis faster and creates reports and predictions after processing the data. It can handle large amounts of both organized (structured) and unorganized (unstructured) data.
- It also helps automate machine learning operations (MLOps), making the process more organized, easy to track, and transparent. This ensures proper control and clear records of all activities.

7.8 Messaging & Notification Services

15. Amazon SNS (Simple Notification Service)

Amazon SNS is a messaging service from AWS that is easy to set up and use. It helps send messages or notifications quickly and at a low cost.

SNS supports two types of messaging:

- Application to Application (A2A): Sends messages between different apps or microservices.
- Application to Person (A2P): Sends messages directly to people through SMS, email, or push notifications.

It works well for bulk messaging, chatting with customers, and connecting different parts of your system. SNS is fully managed, meaning AWS handles the setup and management for you.

16. Amazon SQS (Simple Queue Service)

- Amazon SQS helps different parts of your software send, store, and receive messages safely, even at large volumes, without losing any data. It uses FIFO (First-In-First-Out) to make sure messages are processed in the exact order they arrive and only once.
- SQS allows you to separate and grow microservices, distributed systems, and serverless apps easily. With SQS, messages can be exchanged anytime and anywhere using its message queue system.

7.9 Business & Productivity

17. Amazon Auto-Scaling

Amazon Auto-Scaling is a service that automatically adjusts the number of resources your application needs to keep it running smoothly while saving costs. It can quickly increase or decrease resources, like EC2 instances, depending on the current demand.

There are two types of scaling:

- Dynamic scaling — adjusts to real-time changes in demand.
- Predictive scaling — adjusts based on expected future demand.

Auto-Scaling make sure your application always has the right amount of resources at the right time. It works well with Amazon EC2 Auto Scaling to manage EC2 instances automatically.

18. Amazon ElastiCache

- **Amazon ElastiCache** is a fully-managed AWS service that stores data in memory to make applications and databases run faster. It reduces delays to microseconds, allowing quick access to data with high speed and performance.
- ElastiCache automatically sets up, runs, and scales the data storage in the cloud. It's great for real-time uses like session storage, gaming, caching, and live data analysis.
- It works well with popular open-source caching systems like Redis and Memcached. Since AWS manages it for you, it's easier to use and helps save costs.

19. Amazon Cognito

- Amazon Cognito is an identity management service from AWS. It helps you securely manage and sync data for millions of users.
- With Cognito, users can sign up or log in to your web or mobile apps using their existing accounts like Google, Facebook, Apple, Amazon, or SAML providers. This makes it easy for users while keeping their data secure.
- Cognito has a feature called "user pools" which lets you create user accounts without needing to set up any servers. It also supports strong security, including authentication and data encryption, to protect user information.

7.10 Archival & Backup

20. Amazon S3 Glacier

Amazon S3 Glacier is a low-cost storage service used for long-term data archiving. It is secure, reliable, and allows you to access your data quickly when needed.

S3 Glacier has three storage classes:

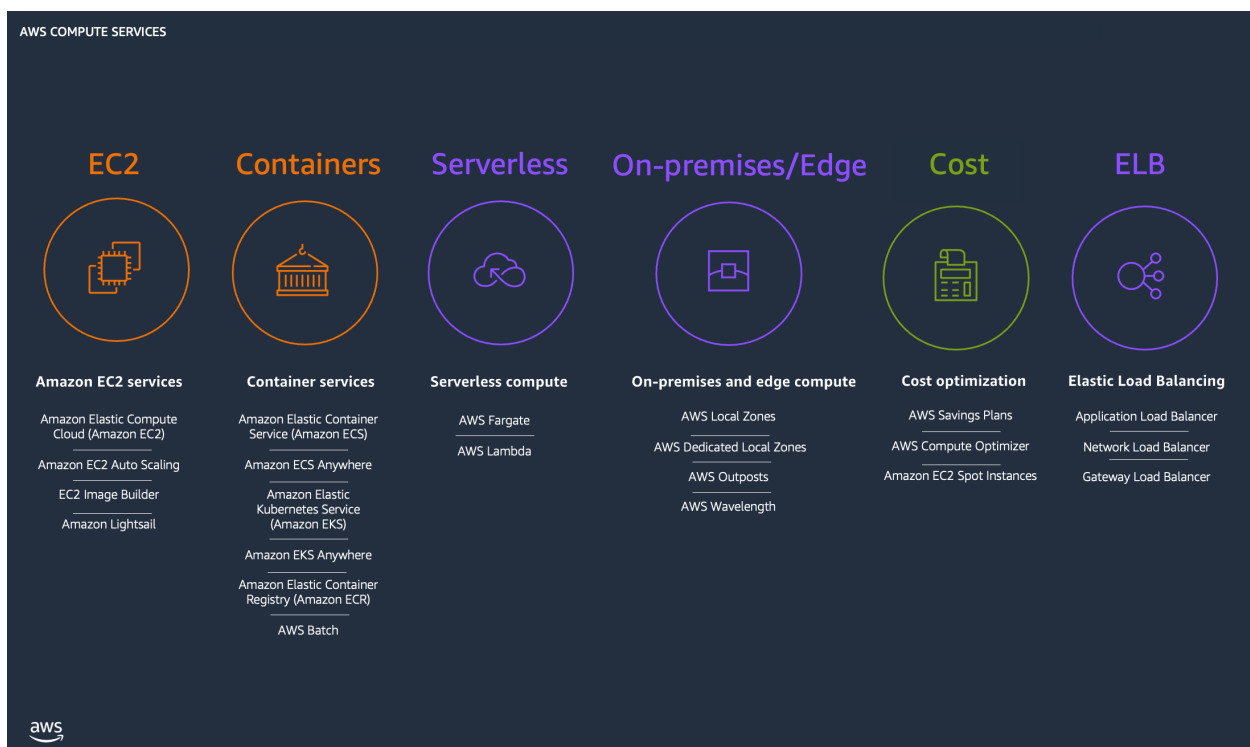
- **Instant Retrieval** gives immediate access to archived data.
- **Flexible Retrieval** allows you to access data when you don't need it right away.
- **Deep Archive** stores data you rarely need, like old records or compliance files.

This service is great for saving money while safely storing large amounts of data for a long time.

7.11 Developer & Serverless Services

21. Amazon LightSail

- **Amazon LightSail** is a Virtual Private Server (VPS) service that helps developers, small businesses, and users easily build and host applications on the cloud.
- It offers everything you need: virtual servers, databases, containers, and storage. LightSail can also work with AWS Lambda to let you create, test, and run code quickly.
- You can set up pre-configured applications with LightSail, using virtual machines, SSD storage, static IP addresses, and data transfer options. It also allows you to easily manage access, networking, and security settings.

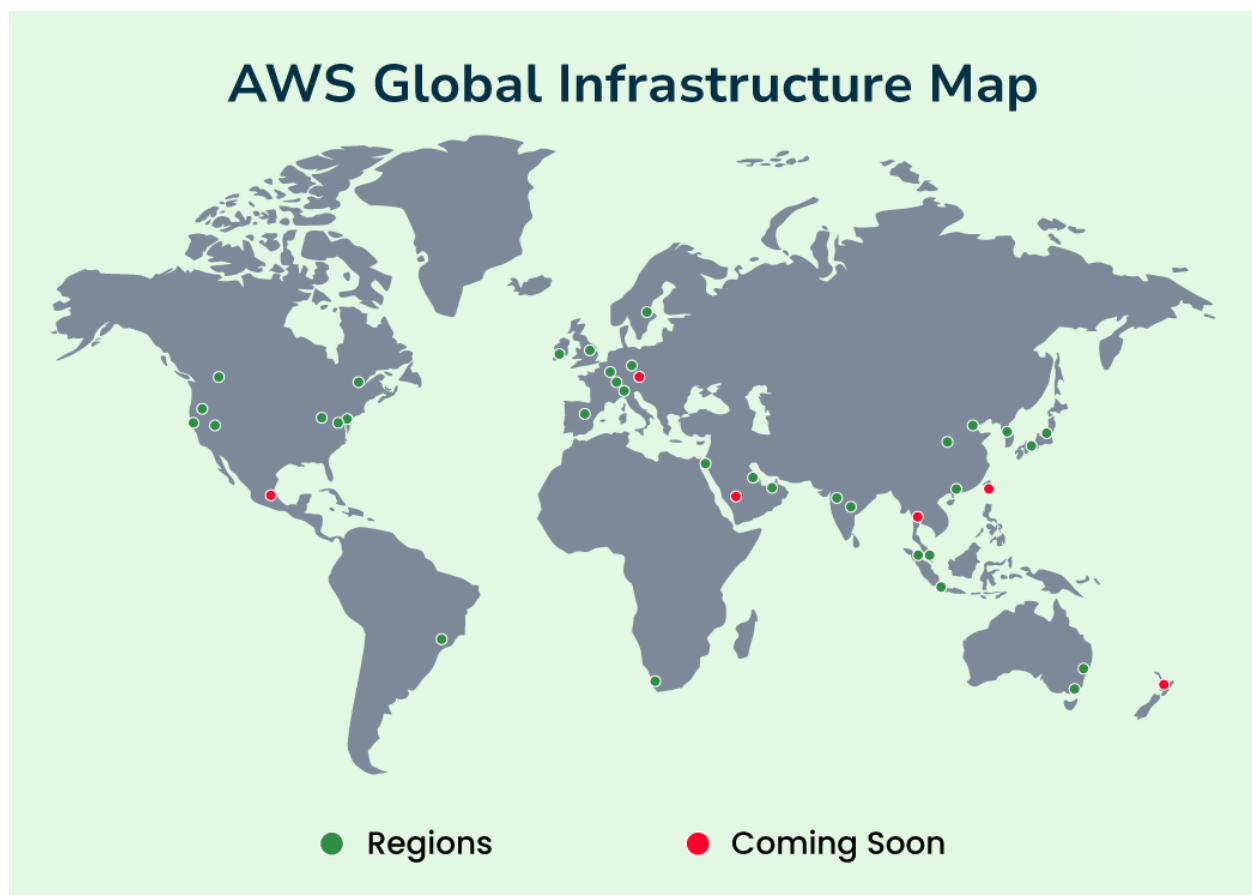


8. AWS Global Infrastructure

The AWS Global Infrastructure Map is a comprehensive visualization of Amazon Web Services expansive global presence. It highlights AWS data centers, including regions, availability zones, and edge locations spread across various geographic areas worldwide.

This infrastructure ensures low-latency access, high availability, and fault tolerance for AWS customers, enabling them to deploy applications and services with greater reliability and performance. By distributing data and services closer to users, AWS global footprint supports scalability and resilience, ensuring that businesses can deliver seamless digital experiences across regions.

The AWS Cloud is distributed across **108 Availability Zones** in **34 regions** worldwide, with future expansions planned for **18 additional Availability Zones** and **six new regions, including Mexico, New Zealand, Saudi Arabia, Thailand, Taiwan, and the AWS European Sovereign Cloud**.



8.1 Need

- AWS Global Cloud Infrastructure is a highly secure, reliable, and expansive cloud platform that offers more than 200 fully featured services from data centers worldwide.
- AWS enables users to deploy application workloads globally in a single step or place applications closer to end users with ultra-low latency, offering single-digit millisecond response times.

- With millions of active customers and a vast network of tens of thousands of global partners, AWS boasts the largest and most vibrant cloud ecosystem.
- AWS is trusted by organizations of all sizes and across nearly every industry, including start-ups, large enterprises, and public sector entities, to run a wide range of use cases.

8.2 Components

1. Data Center

A data center is a physical facility that houses hundreds of computer systems, network devices, and storage appliances. We can run our applications in two or more data centers to achieve high availability, so if there is an outage in one of the data centers, we still have other servers running in another data center. A data center can also deliver cached content to your global end-users to improve response times. At its core, the AWS Global Infrastructure utilizes multiple data centers and group them into Availability Zones, Regions, and Edge Locations.

2. Availability Zone (AZ)

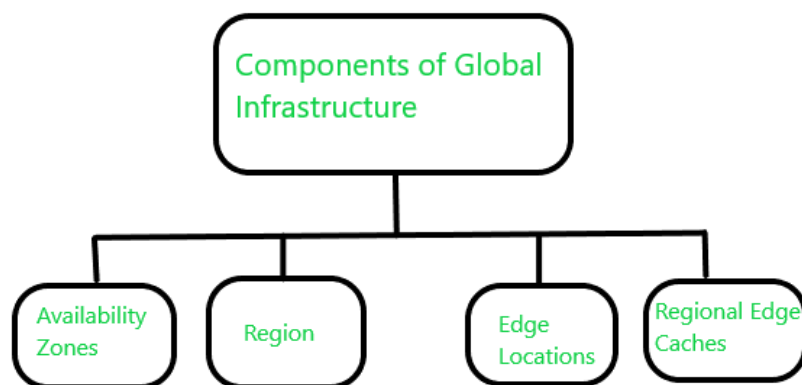
AZs are physically separated data centers with redundant power networking and connectivity. So each AZ is a logical group of data centers which can be one or more physical data centers. They can be in separate buildings or locations. They are built with redundancies. There is high throughput low latency networking between these AZs in a region. All traffic between these AZs is encrypted. Many data centers compose the AWS global infrastructure. Inside the data center, there are thousands of physical servers racks storage and firewalls. Each data center is usually built with redundant power and networking. Each AZ has multiple data centers in each region.

The main reason for having multiple data redundant data centers in an AZ(n) region is for high availability. Many AWS services have built-in high availability, where resources are replicated across multiple AZs in a region. For example, Amazon S3 operates in at least three AZs. Data is protected if one AZ goes down. AWS also gives options to customers to deploy applications in multiple AZs to ensure business continuity in events like a power outage, fires or flood.

3. Point-of-Presence or PoP

The other component of the AWS Global Cloud Infrastructure is the edge networks of Point-of-Presence or PoP. It consists of Edge Locations and Regional Edge Caches, which enables us to distribute our content with low latency to our global users.

Basically, a PoP serves as an access point that allows two different networks to communicate with each other. By using these global edge networks, a user request doesn't need to travel far back to your origin just to fetch data. The cached contents can quickly be retrieved from regional edge caches that are closer to your end users. This is also referred to as a Content Delivery Network or CDN. So for example, we have high-resolution images stored on a server in California. We can cache these media files to an edge location in the Philippines, India, or Singapore to allow our customers in Asia to retrieve these photos faster. The images will be loaded quickly because it is fetched to an edge server near our users, instead of retrieving it from the origin server in California.

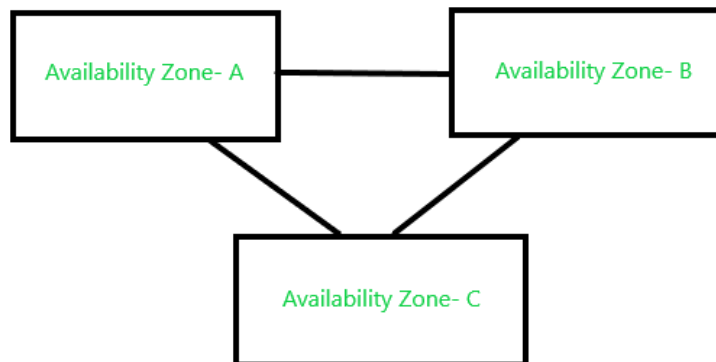


4. Region

When we use AWS console CLI or SDK to manage AWS resources the first thing we need to do is to choose a region. The resources we created in one region are only visible in that region there are a few considerations when we choose a region. First of all, we may want to choose a region that is close to our users for the lowest latency. For example, if the majority of your users are in the US we may want to choose a US region.

The second reason is compliance and regulatory requirements. Certain laws mandate that certain data must be stored in particular countries. For example, if our organization is operating highly sensitive data for the US government we should consider the GovCloud. Some resources or services are only available in certain regions new services are usually launched in the U.S. east northern Virginia region first. Sometimes it takes a long time to roll out to other regions. For example, Alexa for business is only available in this region at the moment however this doesn't mean your end user can't access the application created by the service. It just means we can only create and manage the service in this region each region may have different prices for AWS services. For example, our EC2 instances or data

in S3 buckets may be charged a different price in Singapore than in the US. Keep in mind that AWS charges data transfer between regions.



5. Edge Locations

At the moment, edge locations are part of the AWS Content Delivery Network for low latency high throughput content delivery. Edge locations are all over the world close to the users. They leverage Amazon's ultra-fast global network backbone to deliver data and cache them in a location that is close to the users. Services that use edge locations are Amazon Cloudfront and Lambda Edge Cloudfront is the AWS global CDN for caching dynamic or static content lambda edges the edge computing to run code on low latency computing resources. We only pay as we go with no minimum upfront cost data transfer from AWS origins such as Amazon S3 EC2 and Elastic Load Balancing to the edge location are free. We only pay for the data transferred out of the edge location.

For example, if we are running a photo-sharing website that stores images in an S3 bucket in the US most of our end users are in Singapore, the first time users download images images are delivered from the S3 bucket in the US through the Cloudfront network and cached in the edge location in Singapore. Later on, other users in Singapore will download them from the edge location instead. Without edge locations, these contents will always have to travel from the origin to the end user. We don't need to pay for the data transfer between S3 and the edge location it's much cheaper than sending data from S3 to our users directly.

6. Regional Edge Cache

A Regional Edge Cache in AWS is a location that sits between AWS edge locations and the origin servers (such as S3 or EC2) in the CloudFront content delivery network (CDN). It

helps cache larger objects that may not be frequently accessed at edge locations. When content is not available in an edge location's cache, it is retrieved from the regional edge cache, reducing the need to go back to the origin server, thereby improving content delivery efficiency and reducing latency for users.

Benefits Of AWS Infrastructure

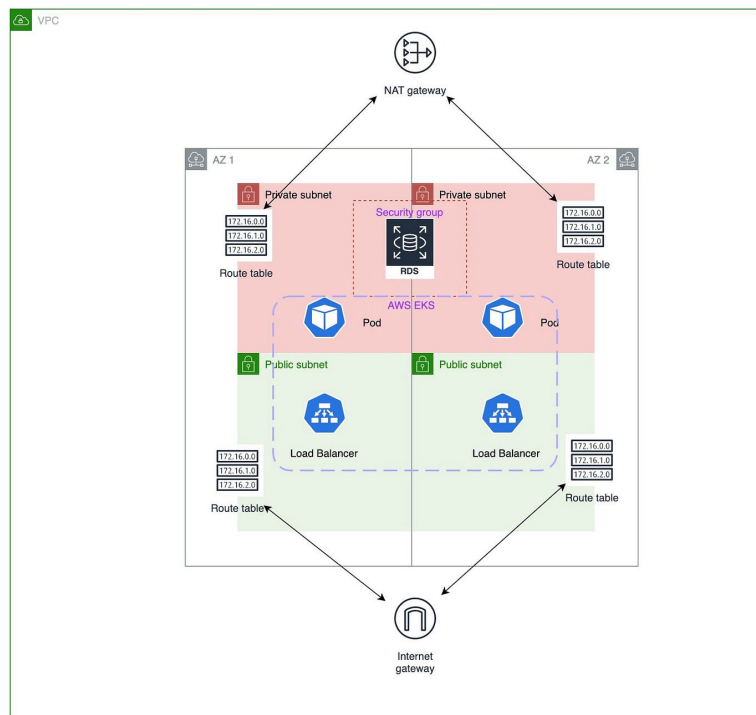
- **Availability:** AWS offers the highest level of network availability among cloud providers. Each AWS region is completely isolated and consists of several Availability Zones (AZs), which are separated sections of infrastructure. By distributing applications across multiple AZs within the same region, users can isolate issues and maintain high availability. AWS's control planes and management consoles are also distributed across regions, and regional API endpoints are designed to remain functional for up to 24 hours, even if isolated from the global control plane, ensuring continuity during any disruptions.
- **Security:** AWS prioritizes security starting with its core infrastructure, which is specifically built for the cloud and adheres to some of the highest global security standards. This infrastructure is constantly monitored, ensuring the confidentiality, integrity, and availability of your data. All data moving across the AWS global network connecting data centers and regions is automatically encrypted at the physical layer before leaving AWS's secure facilities. Users have complete control over their data, with the ability to encrypt, transfer, and manage data retention at any time.
- **Performance:** AWS infrastructure is optimized for high performance. The AWS regions are designed to offer low latency, minimal packet loss, and superior network quality, achieved through a redundant 400 GbE fiber backbone providing terabits of capacity between regions. For applications requiring ultra-low latency, AWS offers Local Zones and AWS Wavelength in partnership with telecommunications providers, bringing infrastructure closer to users and 5G devices. This allows businesses to quickly scale up resources, deploying hundreds or thousands of servers in a matter of minutes to meet varying application needs.
- **Flexibility:** AWS infrastructure offers unmatched flexibility in terms of where and how you run workloads. Whether operating globally across multiple AWS regions and AZs or running workloads with ultra-low latency using AWS Local Zones or AWS Wavelength, the same control plane, APIs, and AWS services are available. If on-premises deployment is preferred, AWS Outposts is an option. For public sector

organizations or highly regulated industries, AWS plans to introduce the AWS European Sovereign Cloud to meet specific regulatory needs.

- **Scalability:** AWS allows businesses to be highly adaptable, utilizing the cloud's scalable infrastructure. Previously, companies had to over-provision to ensure sufficient capacity during peak usage. Now, they can allocate exactly what they need and dynamically scale resources up or down based on demand. This elasticity helps reduce costs while ensuring businesses can respond quickly to changing operational requirements, deploying large numbers of servers in a matter of minutes.
- **Global Footprint:** AWS boasts the largest global infrastructure footprint among cloud providers, which continues to grow rapidly. When deploying workloads, you can select the infrastructure that is closest to your primary users, ensuring the best performance for even the most demanding applications. AWS supports workloads requiring high throughput and low latency, and for unique needs, such as satellite communication, AWS Ground Station provides satellite antennas near AWS regions for seamless data integration from space-based systems.

9. Understanding AWS Network Architecture

AWS network architecture refers to the way AWS services are interconnected to provide secure and efficient connectivity for your applications and resources in the cloud. It's a crucial aspect of cloud computing, impacting scalability, security, and performance. Key components include Virtual Private Clouds (VPCs), subnets, route tables, internet gateways, and various security features.



9.1 Components

- **Virtual Private Cloud (VPC):** A logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network you define. It provides control over your virtual networking environment, including IP address ranges, subnets, route tables, and gateways.
- **Subnets:** Divisions within a VPC that allow you to group resources based on security and access requirements.
 - **Public Subnets:** Subnets with a route to the internet gateway, allowing resources within to communicate with the internet.
 - **Private Subnets:** Subnets without a direct route to the internet, providing a more secure environment for resources like databases.
- **Route Tables:** Determine how network traffic is routed within and outside of a VPC. They contain rules that specify where traffic should be directed based on its destination.
- **Internet Gateway:** A horizontally scaled, redundant, and highly available component that enables communication between your VPC and the internet.
- **Security Groups:** Act as virtual firewalls for your instances, controlling inbound and outbound traffic.
- **Network Access Control Lists (NACLs):** Stateful firewalls that control traffic at the subnet level.
- **VPC Endpoints:** Enable private connections between your VPC and other AWS services, enhancing security by keeping traffic within the AWS network.
- **AWS Direct Connect:** Provides a dedicated network connection between your on-premises network and AWS, improving performance and reducing latency.
- **AWS Transit Gateway:** A service that simplifies the management of complex network topologies, allowing you to connect multiple VPCs and on-premises networks.

9.2 Key Considerations

- **Security:** AWS provides various security features like security groups, NACLs, and VPC endpoints to help protect your resources.

- **Scalability:** AWS networking services are designed to scale with your needs, allowing you to adjust capacity as required.
- **Performance:** Choosing the right networking options and optimizing configurations can significantly impact performance.
- **Cost:** Understanding the cost implications of different networking services is crucial for cost optimization.
- **Hybrid Connectivity:** AWS offers various solutions for connecting your on-premises network with your VPCs, allowing you to leverage both environments.
- **Monitoring and Alerting:** AWS provides tools like CloudWatch and VPC Flow Logs to monitor network traffic and identify potential issues.

By understanding these concepts and components, you can design and implement robust and secure network architectures for your applications in AWS.

10. AWS Identity and Access Management (IAM)

Identity and Access Management (IAM) manages Amazon Web Services (AWS) users and their access to AWS accounts and services. It controls the level of access a user can have over an AWS account & sets users, grant permission, and allows a user to use different features of an AWS account.

Identity and access management is mainly used to manage users, groups, roles, and Access policies. The account we created to sign in to Amazon web services is known as the root account and it holds all the administrative rights and has access to all parts of the account. The new user created an AWS account, by default they have no access to any services in the account & it is done with the help of IAM that the root account holder can implement access policies and grant permission to the user to access certain services.

10.1 Need

We use **IAM** (Identity and Access Management) in AWS to securely manage who can access your resources and what they can do with them.

- 1. User Control:** IAM lets you create users and groups, giving each one specific permissions.
- 2. Security:** It ensures users only have access to what they need, reducing security risks.
- 3. Access Management:** IAM allows you to control who can view or modify resources.

4. Extra Security: You can enable multi-factor authentication (MFA) for added protection.

5.Resource Protection: IAM helps prevent unauthorized changes to your critical infrastructure.

10.2 Working

IAM verifies that a user or service has the necessary authorization to access a particular service in the AWS cloud. We can also use IAM to grant the right level of access to specific users, groups, or services. For example, we can use IAM to enable an EC2 instance to access S3 buckets by requesting fine-grained permissions.



IAM Identities

IAM Identities assists us in controlling which users can access which services and resources in the AWS Console and also we can assign policies to the users, groups, and roles. The IAM Identities can be created by using the Root user

IAM Identities Classified As

1. IAM Users
2. IAM Groups
3. IAM Roles

Root user

The root user will automatically be created and granted unrestricted rights. We can create an admin user with fewer powers to control the entire Amazon account.

IAM Users

We can utilize IAM users to access the AWS Console and their administrative permissions differ from those of the Root user and if we can keep track of their login information.

Example

With the aid of IAM users, we can accomplish our goal of giving a specific person access to every service available in the Amazon dashboard with only a limited set of permissions, such as read-only access. Let's say user-1 is a user that I want to have read-only access to the EC2 instance and no additional permissions, such as create, delete, or update. By creating an IAM user and attaching user-1 to that IAM user, we may allow the user access to the EC2 instance with the required permissions.

IAM Groups

A group is a collection of users, and a single person can be a member of several groups. With the aid of groups, we can manage permissions for many users quickly and efficiently.

Example

Consider two users named user-1 and user-2. If we want to grant user-1 specific permissions, such as the ability to delete, create, and update the auto-calling group only, and if we want to grant user-2 all the necessary permissions to maintain the auto-scaling group as well as the ability to maintain EC2, we can create groups and add this user to them. If a new user is added, we can add that user to the required group with the necessary permissions.

IAM Roles

While policies cannot be directly given to any of the services accessible through the Amazon dashboard, IAM roles are similar to IAM users in that they may be assumed by anybody who requires them. By using roles, we can provide AWS Services access rights to other AWS Services.

Example

Consider Amazon EKS. In order to maintain an autoscaling group, AWS eks needs access to EC2 instances. Since we can't attach policies directly to the eks in this situation, we must

build a role and then attach the necessary policies to that specific role and attach that particular role to EKS.

IAM Policies

IAM Policies can manage access for AWS by attaching them to the IAM Identities or resources IAM policies defines permissions of AWS identities and AWS resources when a user or any resource makes a request to AWS will validate these policies and confirms whether the request to be allowed or to be denied. AWS policies are stored in the form of Jason format the number of policies to be attached to particular IAM identities depends upon no.of permissions required for one IAM identity. IAM identity can have multiple policies attached to them.

IAM Features

Shared Access to your Account: A team working on a project can easily share resources with the help of the shared access feature.

- **Free of cost:** IAM feature of the Aws account is free to use & charges are added only when you access other Amazon web services using IAM users.
- **Have Centralized control over your Aws account:** Any new creation of users, groups, or any form of cancellation that takes place in the Aws account is controlled by you, and you have control over what & how data can be accessed by the user.
- **Grant permission to the user:** As the root account holds administrative rights, the user will be granted permission to access certain services by IAM.
- **Multifactor Authentication:** Additional layer of security is implemented on your account by a third party, a six-digit number that you have to put along with your password when you log into your accounts.

11. AWS Command Line Interface (CLI) Essentials

AWS Command Line (CLI) is a tool designed for managing AWS Services directly from the command line. It provides a lot of options and flexibility compared to the Web Console. It facilitates simplifying the process of interactions with various AWS services such as EC2, Amazon S3, and Amazon EBS providing a consistent command structure. It supports scripting and integration into various development and operational environments making it a versatile tool for AWS Users.

11.1 Working

The AWS Command Line Interface (CLI) is a powerful tool that allows users to interact with AWS services directly from their terminal or command prompt. By providing a unified set of commands, the AWS CLI facilitates efficient management of AWS resources and automation of tasks.

- **Individual Interface:** It provides a consistent command structure for managing various AWS services.
- **Configuration:** AWS users have to configure the CLI with their AWS credentials and default region, enabling secure access to their AWS account.
- **Command Execution:** It allows the execution of commands for a wide range of operations, such as launching instances, managing storage, and monitoring resources.
- **Scripting and Automation:** It supports scripting for automating repetitive tasks and integrating with other tools and services.
- **Output Formatting:** Offers multiple output formats (JSON, text, table) for easy parsing and readability.

Installation

The following are the steps for installing AWS CLI:

Step 1: Download the Installer

- Download the AWS CLI from the AWS CLI Website site:
<https://aws.amazon.com/cli/>
- Choose the type of CLI software based on your OS specifications
- For Linux and macOS users use the following command to download the installation script

```
curl "https://awscli.amazonaws.com/AWSCLIV2.pkg" -o "AWSCLIV2.pkg"
```

Step 2: Run the Installer

- Use the following command to run the installer for the downloaded script in the Linux System:

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
```

- For macOS users run the following command:

```
sudo installer -pkg AWSCLIV2.pkg -target /
```

Step 3: Verify the Installation

- Verify the installation of AWS with the following command:

```
aws --version
```

Step 4: Configure the AWS CLI

- Now, configure with your aws credentials such as Access Key and Secret Key by running the following command:

```
aws configure
```

11.2 Usage

AWS CLI is a software tool that provides control of multiple AWS services directly from the command line. It facilitates simplifying resource management and allows the user to automate the tasks and integrate AWS Services into your workflows efficiently. The following are some of the key points regarding its usage:

- **Resource Management:** Easily manage AWS services like EC2, S3, RDS, and more.
- **Automation:** Script and automate repetitive tasks using shell scripts.
- **Cross-Platform:** Available on Windows, macOS, and Linux, ensuring consistent usage across different environments.
- **Configuration:** Customize your environment with profiles and configuration settings for different use cases.

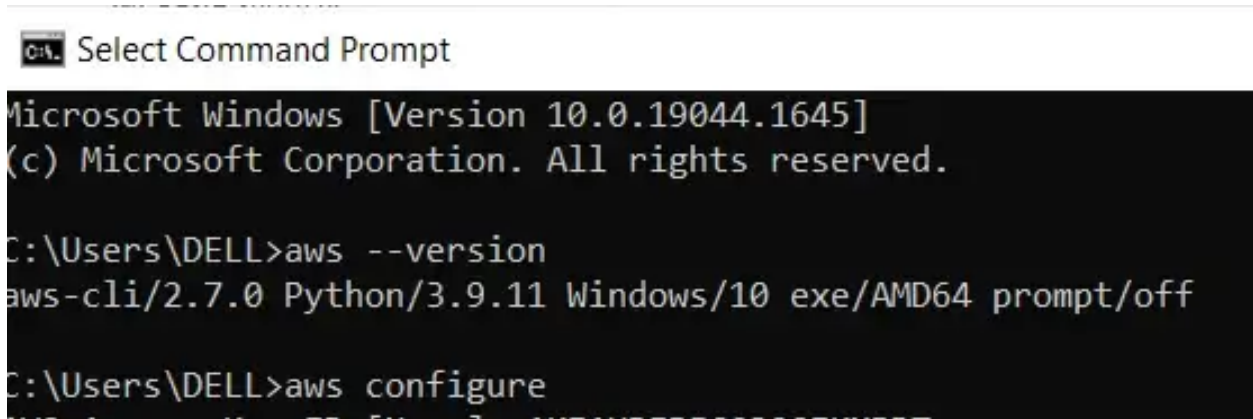
How to use AWS EC2, EBS, and S3 Services in AWS CLI? A Step-By-Step Guide

Step 1: Download the AWS Command Line Interface (CLI) software from AWS Console. With the AWS CLI, you can easily manage your AWS services from the command line, making it a powerful and convenient tool for managing your cloud infrastructure.

- Click on the Windows icon and download the software.
- After installation, open the command prompt and type the command:

aws --version

- It is to verify the installation. This will ensure that the software is properly installed and ready to use.



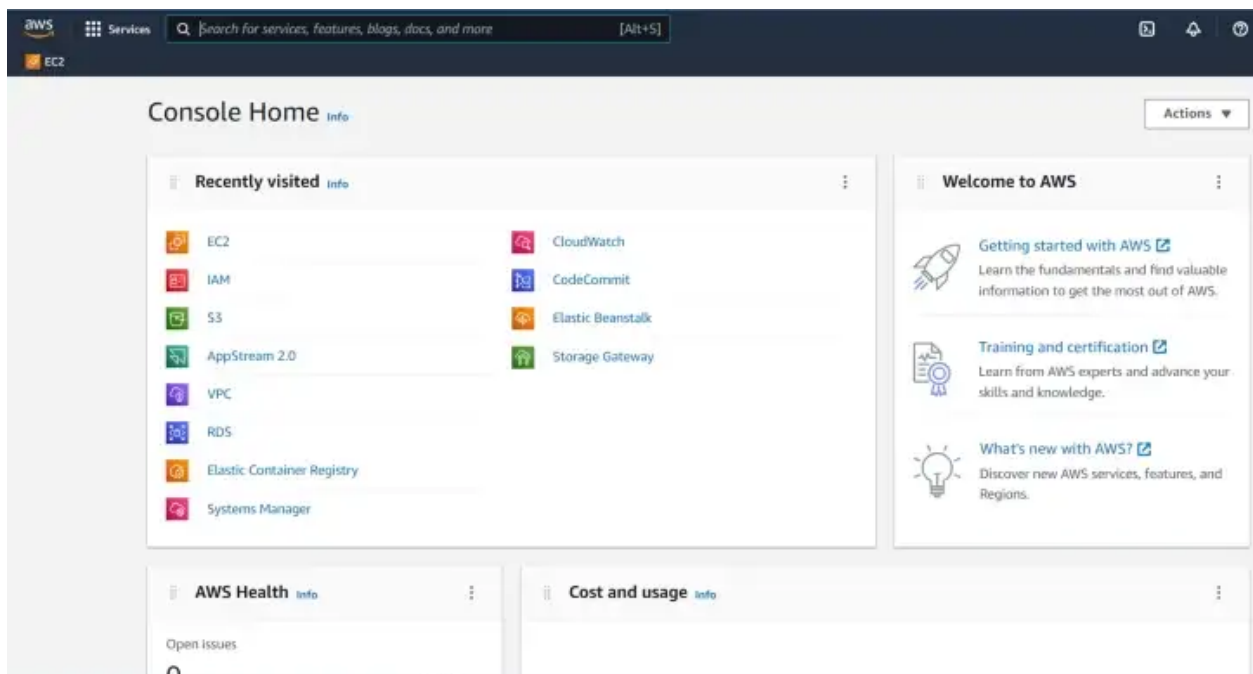
```
Microsoft Windows [Version 10.0.19044.1645]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>aws --version
aws-cli/2.7.0 Python/3.9.11 Windows/10 exe/AMD64 prompt/off

C:\Users\DELL>aws configure
```

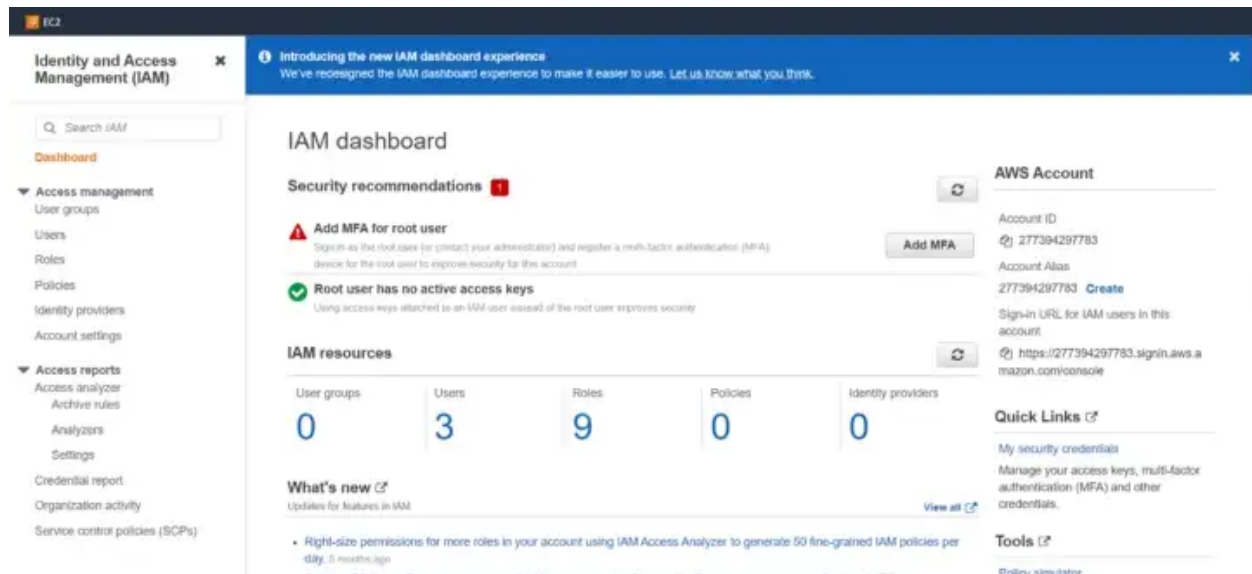
Step 2: Now, we need to configure AWS to take advantage of its powerful cloud computing capabilities. With AWS, you can access a wide range of services, from storage and databases to analytics and machine learning, to help you build and scale your applications.

- Configuring AWS is straightforward, allowing you to quickly get up and running with the cloud.

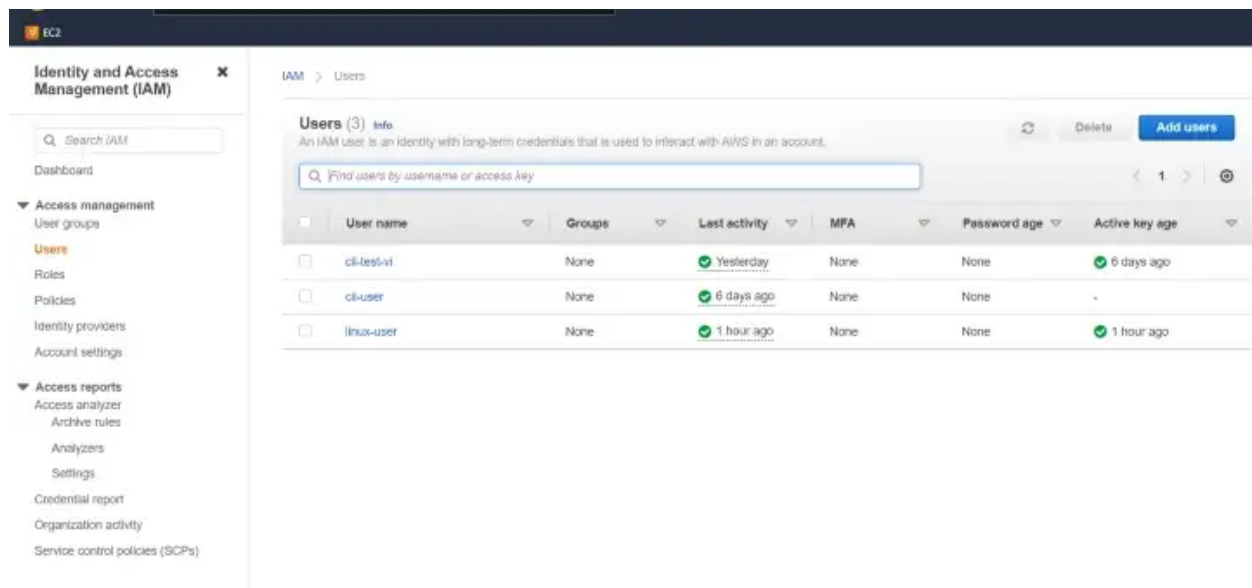


Steps To Create an IAM User in AWS

Step 3: Log in to the AWS console and search for the IAM service.



Step 4: Navigate to the Users tab and Click on the 'Add user' button.



Step 5: Enter a Username. Select an Access Key and click 'Next'.

Set user details

You can add multiple users at once with the same access type and permissions. [Learn more](#)

User name*

[Add another user](#)

Select AWS access type

Select how these users will primarily access AWS. If you choose only programmatic access, it does NOT prevent users from accessing the console using an assumed role. Access keys and autogenerated passwords are provided in the last step. [Learn more](#)

Select AWS credential type* ☒ **Access key - Programmatic access**
Enables an **access key ID** and **secret access key** for the AWS API, CLI, SDK, and other development tools.

☐ **Password - AWS Management Console access**
Enables a **password** that allows users to sign-in to the AWS Management Console.

* Required

[Cancel](#) [Next: Permissions](#)

Step 6: Select the '**Administrator Access**' option and click '**Next**'.

Set permissions

[Add user to group](#) [Copy permissions from existing user](#) [Attach existing policies directly](#)

[Create policy](#)

Filter policies Showing 750 results

	Policy name	Type	Used as
☒	AdministratorAccess	Job function	Permissions policy (3)
☐	AdministratorAccess-Amplify	AWS managed	None
☐	AdministratorAccess-AWSElasticBeanstalk	AWS managed	None
☐	AlexaForBusinessDeviceSetup	AWS managed	None
☐	AlexaForBusinessFullAccess	AWS managed	None
☐	AlexaForBusinessGatewayExecution	AWS managed	None
☐	AlexaForBusinessLifesizeDelegatedAccessPolicy	AWS managed	None
☐	AlexaForBusinessPolyDelegatedAccessPolicy	AWS managed	None
☐	AlexaForBusinessReadOnlyAccess	AWS managed	None

[Cancel](#) [Previous](#) [Next: Tags](#)

Step 7: Click on the '**Create user**' button.

EC2

Review

Review your choices. After you create the user, you can view and download the autogenerated password and access key.

User details

User name	awscli-blog
AWS access type	Programmatic access - with an access key
Permissions boundary	Permissions boundary is not set

Permissions summary

The following policies will be attached to the user shown above.

Type	Name
Managed policy	AdministratorAccess

Tags

The new user will receive the following tag

Key	Value
shubhi94	{empty}

[Cancel](#)
[Previous](#)
[Create user](#)

- The following screenshot shows the successful creation of an IAM user.

EC2

Add user

1 2 3 4 5

Success

You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.

Users with AWS Management Console access can sign-in at: <https://277394297783.signin.aws.amazon.com/console>

[Download .csv](#)

User	Access key ID	Secret access key
------	---------------	-------------------

Configure AWS CLI With Credentials

Step 8: Log in to AWS CLI using your Access Key ID and Secret Access Key for secure access to your account. With these credentials, you can access your account and take advantage of the powerful features of the AWS Command Line Interface.

- Download the .csv File by following these Steps:
- Open the command line and type in the command
aws configure
- Copy and paste your IAM user Access Key ID and Secret Access Key into the command line.

- Enter the region name where you would like to launch your instance.
- Select the output format, such as JSON.

```
C:\Users\DELL>aws configure
AWS Access Key ID [*****T7JX]: AKI
AWS Secret Access Key [*****ojsS]:
Default region name [ap-south-1]: ap-south-1
Default output format [[Json]]: json
```

Creating EC2 Instances

Step 9: Launch an Amazon EC2 Instance with AWS CLI.

- Utilizing the AWS Command Line Interface (CLI), you can quickly and easily launch an Amazon EC2 instance, allowing you to access the power of the cloud with just a few simple commands.
- With the AWS CLI, you can quickly and easily spin up a virtual machine in the cloud, giving you the ability to access the power of the cloud with ease.
- We will launch an EC2 instance with the help of this command. You can select any image ID from the AWS console, use an existing key pair, select any existing security group ID, and use any existing subnet ID.
- All of these IDs must belong to the same availability zone for the instance to be launched successfully.

```
C:\Users\DELL>aws ec2 run-instances --image-id ami-079b5e5b3971bd10d --count 1 --instance-type t2.micro --key-name shubhi94 --security-group-ids sg-0192780edb580d7a3 --subnet-id subnet-07332f2ba537dd1dc
{
  "Groups": [],
  "Instances": [
    {
      "AmiLaunchIndex": 0,
      "ImageId": "ami-079b5e5b3971bd10d",
      "InstanceId": "i-05c7e636eaff74594",
      "InstanceType": "t2.micro",
      "KeyName": "shubhi94",
      "LaunchTime": "2022-05-18T19:18:20+00:00",
      "Monitoring": {
        "State": "disabled"
      },
      "Placement": {
        "AvailabilityZone": "ap-south-1a",
        "GroupName": "",
        "Tenancy": "default"
      },
      "PrivateDnsName": "ip-172-31-10-38.ap-south-1.compute.internal",
      "PrivateIpAddress": "172.31.10.38",
      "ProductCodes": [],
      "PublicDnsName": "",
      "State": {
        "Code": 0,
        "Name": "pending"
      },
      "StateTransitionReason": "",
      "SubnetId": "subnet-07332f2ba537dd1dc",
      "VpcId": "vpc-0be5965e9c47e6f53",
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "55865310-92ec-48f0-9b09-063948fb3f00",
    }
  ]
}
```

- ## Creating Key Pair Using AWS CLI

- With the CLI, you can quickly and easily generate a key pair that will provide you with the highest level of security for your data.
- Create a key pair using this command. This command will enable you to securely generate a unique set of cryptographic keys that can be used to authenticate and encrypt data. With this key pair, you can ensure that your data is kept safe and secure.

```
aws ec2 create-key-pair --key-name vivek1
```

- Provide the name of the key pair to gain access.

- Now successfully created, this accomplishment marks a milestone in our journey toward success. We have worked hard to reach this point, and we are proud of the progress we have made.
- This achievement is a testament to our dedication and commitment to excellence.

Creating EBS Volume using AWS CLI

Step 11: Utilizing the AWS Command Line Interface (CLI), we will create an Elastic Block Store (EBS) Volume to store our data. This will provide us with a secure, reliable, and cost-effective storage solution for our data.

- Using this command, you can create an EBS Volume. This powerful tool allows you to quickly and easily create a reliable storage solution for your data. With EBS Volume, you can store your data securely and access it whenever you need it.
- Plus, you can scale up or down as needed, giving you the flexibility to meet your changing storage needs.

```
C:\Users\DELL>aws ec2 create-volume --volume-type gp2 --size 8 --availability-zone ap-south-1a
{
  "AvailabilityZone": "ap-south-1a",
  "CreateTime": "2022-05-18T19:30:52+00:00",
  "Encrypted": false,
  "Size": 8,
  "SnapshotId": "",
  "State": "creating",
  "VolumeId": "vol-0cd6198dcdbb8a04e",
  "Iops": 100,
  "Tags": [],
  "VolumeType": "gp2",
  "MultiAttachEnabled": false
}
```

- Now Successfully Created.

Attaching EBS Volume to EC2 Instance Using AWS CLI

Step 12: We will attach EBS volume to the EC2 Instance using AWS CLI.

- Using this command attach EBS Volume to EC2 Instance.

```
aws ec2 attach-volume
--volume-id vol-1234567890abcdef0
--instance-id i-01474ef662b89480
--device /dev/sdf
```

```
C:\Users\DELL>aws ec2 attach-volume --volume-id vol-0cd6198dcdbb8a04e --instance-id i-07e7e704e8f948b40 --device /dev/sdf
```

Adding Security Groups Using AWS CLI

Step 13: We will create a Security Group using AWS CLI. Use the following command:

```
aws ec2 create-security-group --group-name my-sg --description "My sg group" --vpc-id
vpc-f0e73c9b
```

- **VPC ID:** To find the [VPC](#) ID, navigate to the VPC service in the Amazon EC2 console and click on VPCs.
- Now, we have successfully created a security group in Amazon EC2, allowing us to secure our resources and protect our data.

```
aws> ec2 create-security-group --group-name my-sg --description "my sg group" --vpc-id vpc-f0e73c9b
-----
|          CreateSecurityGroup          |
+-----+-----+
| GroupId | sg-0ad8f4b8de242b1e3 |
+-----+-----+
aws> █
```

Step 14: Utilizing the AWS Command Line Interface (CLI), we will create an S3 Bucket to store our data. This bucket will provide a secure and reliable storage solution for our data, ensuring that it is always accessible and protected.

- To upload an object to an S3 bucket, use the command:

```
aws s3 cp object_name s3://bucketname
```

```
An error occurred (InvalidVolume.ZoneMismatch) when calling the AttachVolume operation: The volume 'vol-04f535e8b073eb475' is not in the same
availability zone as instance 'i-0bb4fcf012fbc0b05'
root@vivek:/home/vivek# aws s3 mb s3://vivekdis
make_bucket: vivekdis
root@vivek:/home/vivek# aws s3 cp /home/vivek/workspce/red.png s3://vivekdis/
```

- Then, use this command:

```
aws s3api put-object-acl --bucket (bucketname) --key (object-name) --acl-public-read
```

- Now, the object is accessible to the public.

Step 15: After you the successfully done all the steps you can see your object in your web browser using the object URL.



11.3 AWS CLI Commands

The following are some of the AWS CLI Commands:

1. Configure AWS CLI

- The following command is used for configuring AWS CLI with credentials such as access key, secret key, and output format.

```
aws configure
```

2. List Amazon S3 Buckets

The following is used for listing Amazon S3 Buckets that are available in your AWS Account:

```
aws s3 ls
```

3. Copy Files to Amazon S3

- The following command is used for copying the file on your local system to an Amazon S3 bucket.

```
aws s3 cp localfile.txt s3://bucketname/
```

4. Creating an Amazon S3 Bucket

The following command is used to create an Amazon S3 Bucket:

```
aws s3 mb s3://mybucket
```

5. Delete an Amazon S3 Bucket

The following command is used to delete an Amazon S3 Bucket:

```
aws s3 rb s3://mybucket --force
```

6. List Amazon EC2 Instances

- The following command is used for listing the Amazon EC2 instances by describing them:

```
aws ec2 describe-instances
```

7. Start an Amazon EC2 Instance

- The following command is used to start an Amazon EC2 Instance by specifying its instance ID:

```
aws ec2 start-instances --instance-ids i-1234567890abcdef0
```

8. Stop an Amazon EC2 Instance

- The following command is used to stop an Amazon EC2 Instance by specifying the respective instance ID:

```
aws ec2 stop-instances --instance-ids i-1234567890abcdef0
```

9. List Lambda Functions

- The following command is used to list the lambda functions:

```
aws lambda list-functions
```

10. Invoke a Lambda Function

The following command is used to invoke a AWS lambda function and it outputs the response to a file.

```
aws lambda invoke --function-name myfunction out.txt
```

11. Create A DynamoDB Table

The following command is used to create a DynamoDB table with specifying the attributes and provisioning the throughput.

```
aws dynamodb create-table --table-name mytable --attribute-definitions  
AttributeName=Id,AttributeType=S --key-schema AttributeName=Id,KeyType=HASH --  
provisioned-throughput  
ReadCapacityUnits=1,WriteCapacityUnits=1
```

12. List CloudFormation Stacks

- The following command is used to list the CloudFormation stacks:

```
aws cloudformation list-stacks
```

13. Deploy CloudFormation Stack

- The following command is used for deploying a CloudFormation stack with the template.yaml file that is defined with configurations.

```
aws cloudformation deploy --template-file template.yaml --stack-name my-stack --  
capabilities CAPABILITY_IAM
```

14. List All RDS Instances

- The following command is used for all the Amazon RDS instances:

aws rds describe-db-instances

11.4 Features

The following are the features of AWS CLI:

1. **Unified Interface:** It provides a consistent command structure for managing various AWS services.
2. **Automation:** It allows scripting and automation of repetitive tasks while enhancing efficiency and consistency.
3. **Cross-Platform Compatibility:** It facilitates to work seamlessly on Windows, macOS, and Linux, ensuring flexibility across different environments.
4. **Configuration Management:** It comes up with supporting the profiles and configuration settings for customizing environments and managing multiple accounts.
5. **Output Formats:** It offers multiple output formats (JSON, text, table) for easy parsing and readability for fulfilling various types of needs.

11.5 Benefits

The following are the benefits of AWS CLI:

1. **Streamlined Management:** AWS CLI provides a unified command line tool for managing multiple AWS services, simplifying resource management tasks.
2. **Automation:** It enables scripting and automation of repetitive tasks, allowing for efficient batch operations and workflow automation.
3. **Cross-Platform Compatibility:** It works seamlessly across different operating systems, including Windows, macOS, and Linux, ensuring consistent usage in diverse environments.
4. **Customization:** It provides the support of configuring profiles and settings, allowing users to customize their environment and manage multiple AWS accounts with ease.
5. **Integration:** AWS CLI will easily integrate with other tools and services facilitating seamless integration into existing workflows and development pipelines.